```
D:\Software\Python\Python311\python.exe E:\PythonProjectsByYear\Year2025\MC-PIKAN_Official\experiments\Volterra1D_MCPI.py
The network model is cPIKAN
Layers: ModuleList(
  (0): ChebyKANLayer(in_features=1, out_features=10, degree=6)
  (1): ChebyKANLayer(in_features=10, out_features=10, degree=6)
  (2): ChebyKANLayer(in_features=10, out_features=1, degree=6)
)
Total trainable parameters: 840
Model on device: cuda:0
Start training...
Max Epochs: 2000, Learning Rate: 0.001, Epsilon: 1e-20, Optimizer: Adam
Training: 100%|██████████| 2000/2000 [00:08<00:00, 223.73epoch/s, Loss=1.93e-06, LR=1.0e-03, Best=1.93e-06]
Training finished. Best loss: 1.30e-06. Model saved to results/best_model.pth
Loss history saved to results/loss_history.npy
Training time: 9.730039119720459s
Relative loss: 0.0017865969566628337
L2 loss: 0.001136814127676189
Parameter containing:
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        [[ 2.5325e-02,  3.4688e-02,  5.3248e-03, -8.7550e-03,  2.1872e-02,
           1.3269e-02,  3.9677e-02]],

        [[ 8.9211e-03, -3.9193e-02, -2.2070e-02,  1.4060e-02,  2.5399e-02,
          -4.4074e-02, -3.0133e-02]],
```

```

        [[-1.5876e-02,  1.0415e-02, -2.0999e-02, -7.3541e-03,  3.6677e-02,
          -6.0140e-03, -4.2536e-02]],

        [[ 2.4456e-02, -5.6268e-02,  1.5242e-02,  3.4449e-02, -2.0776e-03,
          -4.1596e-03,  1.7123e-02]]], device='cuda:0', requires_grad=True)
==========================================================================================
Layer (type:depth-idx)                   Output Shape              Param #
==========================================================================================
CPIKANModel                              [1, 1]                    --
├─ModuleList: 1-1                          --                        --
│    └─0.weight                                                        ├─70
│    └─1.weight                                                        ├─700
│    └─2.weight                                                        └─70
│    └─ChebyKANLayer: 2-1                   [1, 10]                   70
│    │    └─weight                                                      └─70
│    └─ChebyKANLayer: 2-2                   [1, 10]                   700
│    │    └─weight                                                      └─700
│    └─ChebyKANLayer: 2-3                   [1, 1]                    70
│    │    └─weight                                                      └─70
==========================================================================================
Total params: 840
Trainable params: 840
Non-trainable params: 0
Total mult-adds (Units.MEGABYTES): 0.00
==========================================================================================
Input size (MB): 0.00
Forward/backward pass size (MB): 0.00
Params size (MB): 0.00
Estimated Total Size (MB): 0.00
==========================================================================================

进程已结束，退出代码为 0

```